RAJALAKSHMI ENGINEERING COLLEGE

Ajay Preethan K 251801010-2025-AIDS

A2

Started on	Thursday, 8 October 2026, 10:06 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:11 AM
Time taken	5 mins 41 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    long long ways[n + 1];

    ways[0] = 1;

    for (int i = 1; i <= n; i++)
    {
        ways[i] = ways[i - 1];

        if (i >= 3)
        {
            ways[i] = ways[i] + ways[i - 3];
        }
    }

    printf("%lld", ways[n]);

    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Back to Course